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HOTEL BOOKING SYTEM

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CHAPTER ONE INTRODUCTION

1.1 Background of the Study

The rapid advancement of information technology, particularly mobile applications and internet-based services, has significantly transformed how people access services such as travel and accommodation. Hotel booking systems have become essential tools that enable users to search, compare, and reserve hotel rooms efficiently and conveniently.

In many developing regions, including Somalia, hotel booking is still largely conducted through traditional methods such as phone calls, physical visits, and manual record-keeping. These methods are inefficient, time-consuming, and prone to errors such as double booking and inaccurate data.

Therefore, there is a need for a modern, automated, and reliable solution that improves efficiency and enhances the user experience. This study aims to develop a Hotel Booking System using PHP and MySQL to provide a fast, secure, and easy-to-use system.

1.2 Statement of the Problem

Despite the availability of many hotels, customers still face challenges when booking rooms. These challenges include a lack of accurate and up-to-date information, time-consuming booking procedures, and the risk of booking errors such as double reservations.

Additionally, there is no centralized application that allows users to easily search, compare, and book hotels in one place. In regions like Somalia, the absence of efficient digital platform further complicates the booking process and reduces service quality.

Therefore, there is a need to develop a reliable and user-friendly hotel booking system that addresses these issues and improves overall service delivery.

1.3 Objectives of the Study

1.3.1 General Objective

The main objective of this project is to design and develop a hotel booking system that simplifies and improves the process of searching and booking hotel rooms.

1.3.2 Specific Objectives

- To develop a user-friendly platform interface
- To provide hotel search and filtering features
- To implement a secure booking system
- To store and manage user and booking data
- To include a review and rating application
- To provide booking confirmation notifications

1.4 Feasibility Study

1.4.1 Technical Feasibility

The project is technically feasible because the required technologies such as PHP and MySQL are readily available and widely used. The developers also have the necessary skills to implement the system successfully.

1.4.2 Economic Feasibility

The cost of developing the system is relatively low since open-source tools are used. Additionally, the system has the potential to generate economic benefits by increasing hotel bookings and customer satisfaction.

1.4.3 Operational Feasibility

The system is easy to use and accessible, making it suitable to a wide range of users. It can be operated on mobile devices, allowing users to book hotels anytime and anywhere.

1.4.4 Schedule Feasibility

The project can be completed within the allocated academic timeframe. Proper planning and scheduling will ensure timely completion of all development phases.

1.5 Scope of the Study

The system will focus on providing core functionalities such as user registration and login, hotel search and filtering, viewing hotel details including price, location, and reviews, and booking hotel rooms. It will also allow users to manage their profiles and provide feedback through a review and rating system.

However, the system will not include advanced features such as flight booking or full online payment integration due to time and resource limitations.

1.6 Significance of the Study

This project is significant because it provides users with a convenient way to book hotel rooms quickly and efficiently. It also helps hotel owners reach more customers and improve their services. Furthermore, it offers developers practical experience in application design and development.

1.7 Target Beneficiaries

The primary beneficiaries of this project include travelers, hotel owners, tourism companies, and students interested in system development and research.

1.8 Methodology

1.8.1 Data Collection Methods

Data for this project will be collected through interviews, observation, and document review.

1.8.2 System Development Process

The development process will include problem identification, requirement analysis, system design, development, testing, and implementation.

1.8.3 Development Model

The Agile (Iterative) model will be used to allow continuous improvement and flexibility during system development.

1.9 Limitations of the Study

The system depends on internet connectivity, which may limit access in some areas. Furthermore, payment integration may not be fully implemented, and real-time updates may occasionally experience delays.

CHAPTER TWO

REQUIREMENT ANALYSIS AND SPECIFICATION

2.1 Introduction

This chapter presents the system analysis and requirements needed to develop the Hotel Booking Management System. It describes the existing system, identifies its limitations, and presents the proposed system as a solution. In addition, it outlines both functional and non-functional requirements of the new system.

2.2 Existing System

Currently, many hotels rely on manual or partially computerized systems to manage bookings and customer information. Customers typically reserve rooms through phone calls, physical visits, or paper-based records.

These traditional methods are inefficient and often lead to delays, data loss, and inaccurate information. Since there is no centralized system, managing bookings and tracking room availability becomes difficult, especially during peak seasons.

2.3 Participants in the Existing System

The current system involves the following participants:

Customer: A person who wants to book a room

Staff: Responsible for recording bookings

Hotel Manager: Oversees the overall operations

These users rely on a manual system, which results in inefficiencies and errors.

2.4 Major Functions of the Existing System

The existing system performs basic operations such as recording customer bookings, manually assigning rooms, handling payments, and maintaining records using paper-based methods. These functions are carried out without automation, which reduces efficiency and increases the risk of errors.

2.5 Problems of the Existing System

The existing system faces several major challenges. It is time-consuming and prone to human error due to manual data handling. There is no reliable database to store and manage information, making it difficult to retrieve records when needed.

Additionally, the system does not provide real-time updates, and customers are unable to easily check room availability. The absence of an online platform limits accessibility and reduces customer satisfaction. These issues lead to poor service delivery and overall customer dissatisfaction.

2.6 Proposed System

To address these challenges, a Hotel Booking Management System is proposed. This system provides an automated and user-friendly platform for managing hotel operations.

The proposed system will allow users to search for available rooms, book rooms, and process payments efficiently. It will also enable administrators to manage rooms, staff, and reports using a centralized database.

By using technologies such as PHP and MySQL, the system will ensure data accuracy, improve performance, and provide real-time access to information.

2.7 Requirements of the Proposed System

2.7.1 Functional Requirements

The Hotel Booking Management System shall provide the following functionalities:

- Manage rooms
- Manage room types and pricing
- Manage bookings
- Process and manage payments
- Search available rooms
- Filter rooms based on user preferences

- Manage staff information
- Generate system reports
- Display dashboard information
- Authenticate users, including login, registration, and password management

2.7.2 Non-Functional Requirements

The system should meet the following quality requirements:

Security: Ensure authentication and authorization

Performance: Provide fast response time

Usability: Ensure user-friendly interface

Reliability: Ensure system stability

Scalability: Support system growth

2.8 System Analysis

This section presents the modeling of both the existing and proposed systems using appropriate analysis techniques. These models help improve understanding of system requirements and how different components interact. Each model is directly related to the System Requirement Specification (SRS) and provides a clear representation of system functionality and behavior.

2.8.1 System Requirement Specification (SRS)

The System Requirement Specification defines the functional and non-functional requirements of the Hotel Booking Management System. These requirements are modeled using different diagrams to provide a clear understanding of the system processes and structure.

2.8.1.1 Use Case Model Diagram

The use case diagram represents the interaction between users (actors) and the system. It identifies the main functionalities that each user can perform.

Narrative Description

The main actors in the system are Customer, Admin, and Staff. The Customer can register, log in, search for rooms, make bookings, and perform payments. The admin manages rooms, staff, and reports, while Staff assists in booking management.

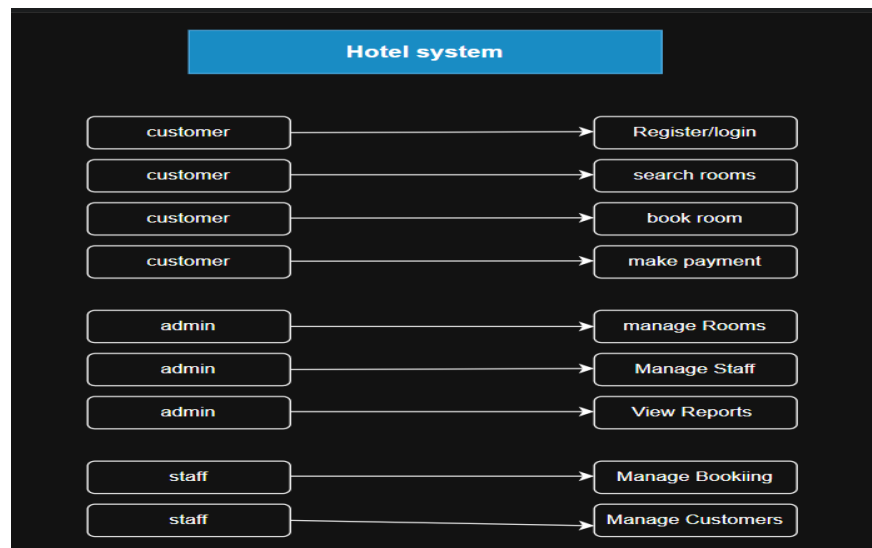


Figure 2.1 Use Case Model Diagram

2.8.1.2 Use Case Documentation

Use Case 1: User Registration

Actor: Customer

Description: The user creates a new account

Pre-condition: User does not have an account

Post-condition: User account is created successfully

Use Case 2: User Login (Security Login)

Actor: Customer / Admin / Staff

Description: The user logs into the system securely

Pre-condition: User must have a registered account

Post-condition: The user gains access to the system

Use Case 3: Book Room

Actor: Customer

Description: The user books an available room

Pre-condition: User must be logged in

Post-condition: The booking is stored in the system

Use Case 4: Make Payment

Actor: Customer

Description: The user completes payment for a booking

Pre-condition: Booking must exist

Post-condition: Payment is recorded

2.8.1.3 Sequence Diagram

The sequence diagram shows how different components interact over time during the booking process.

Narrative Description:

The user sends a request to log in, which is verified by the system. After login, the user searches for rooms, and the system retrieves available room data from the database. The user selects a room and submits a booking request. The system processes the request, stores the booking, and confirms it. Finally, the user completes payment, and the system updates the payment status.

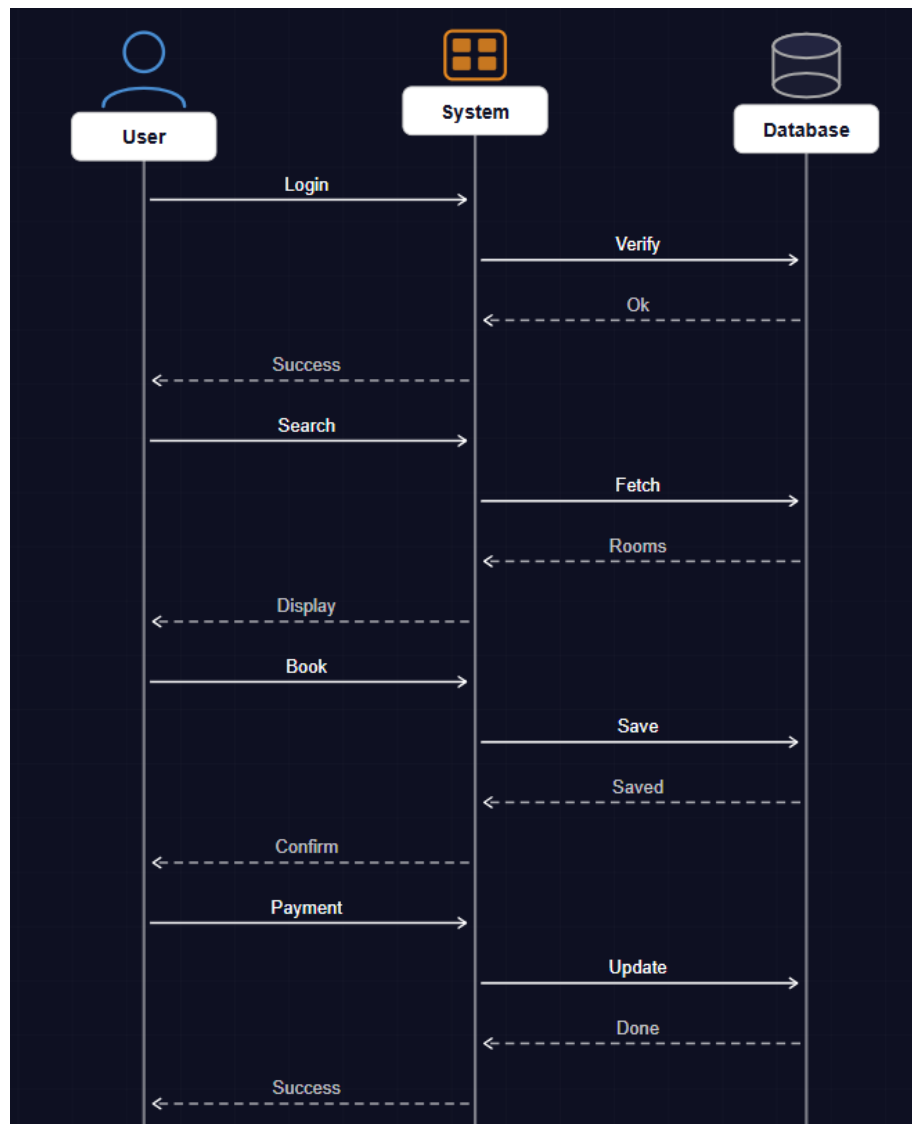


Figure 2.2 Sequence Diagram for Booking Process

2.8.1.4 State Chart Diagram

The state chart diagram represents different states of the booking process and how the system transitions between them.

Narrative Description:

A booking transitions through several states such as “Pending,” “Confirmed,” and “Completed.” Initially, when a booking is created, it is in a pending state. After payment is made, the booking is confirmed. Once the service is completed, the booking status changes to completed.

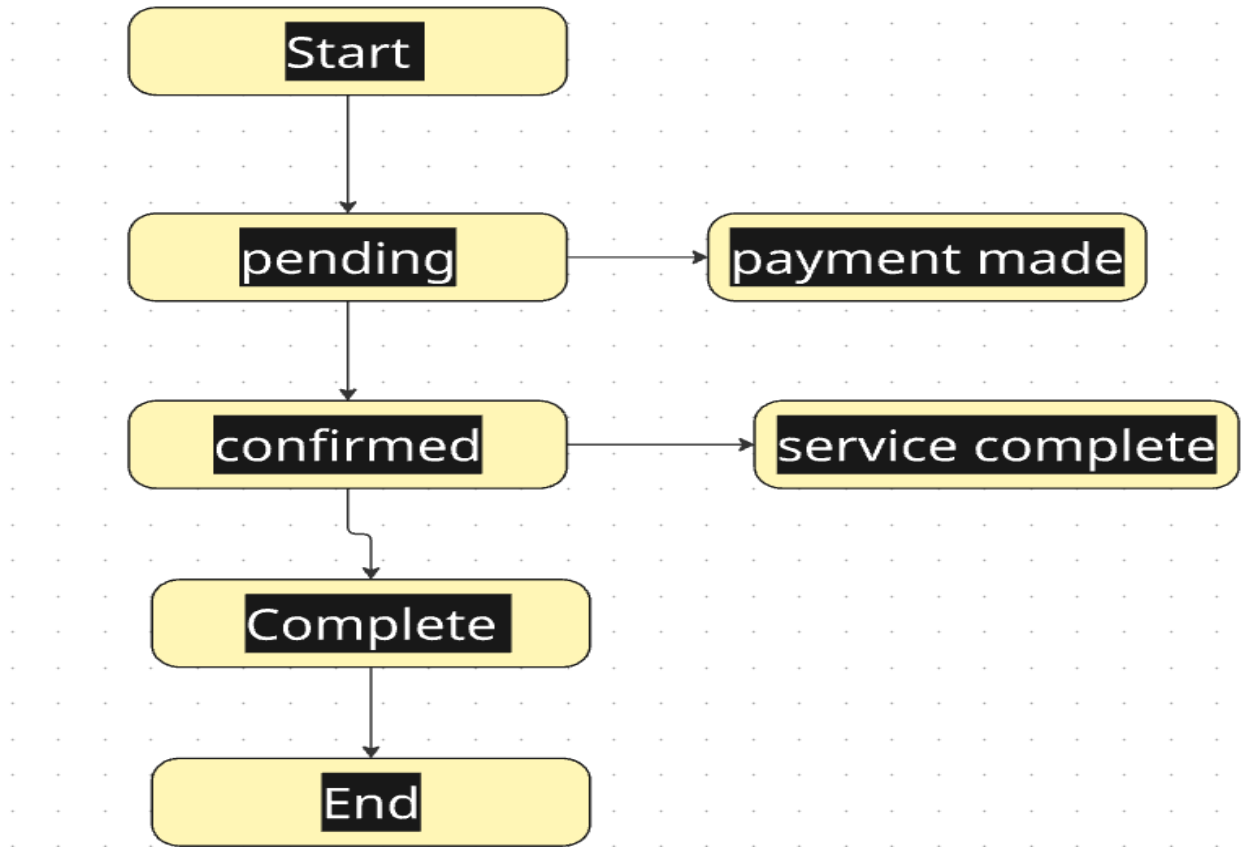


Figure 2.3 State Chart Diagram of Booking Process

2.8.1.5 Activity Diagram

The activity diagram shows the workflow of processes within the system.

Narrative Description:

The process begins when the user logs into the system. The user searches for available rooms and selects one. The system checks availability and allows the user to proceed with booking. After entering details, the user confirms the booking and completes payment. The system then records the transaction and terminates the process.

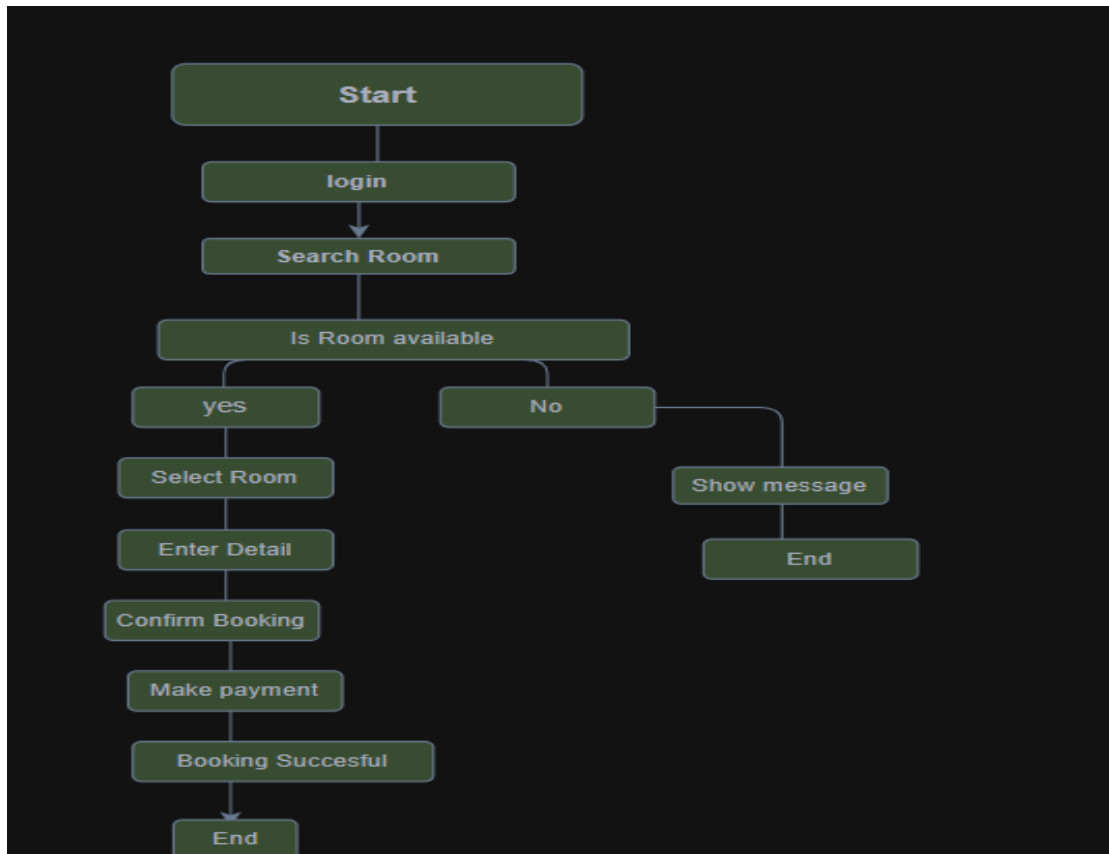


Figure 2.4 Activity Diagram of Booking Workflow

2.8.1.6 Class Diagram

The class diagram represents the structure of the system, including classes, attributes, and relationships.

Narrative Description:

The main classes in the system include Customer, Room, Booking, and Payment. Each class contains attributes such as customer details, room information, booking status, and payment details. Relationships between these classes define how data is managed within the system.

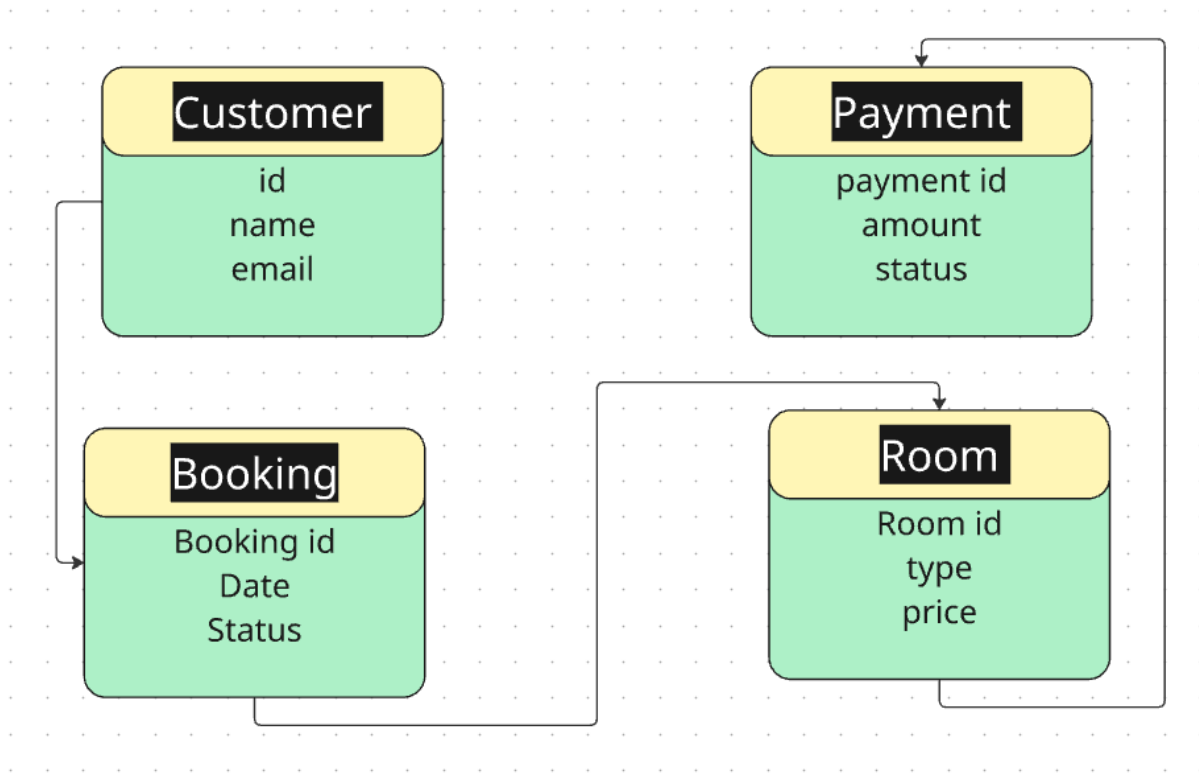


Figure 2.5 Class Diagram of the System

2.8.1.7 User Interface Prototyping

User interface prototyping provides a visual representation of how the system will look and how users will interact with it.

Narrative Description:

The system interface includes several screens such as login page, registration page, dashboard, room listing page, booking form, and payment page. These interfaces are designed to be simple, user-friendly, and easy-to-navigate.

Login

Enter your credentials to access the hotel booking system

Email

Enter your email

Password

Enter your password

Login

Figure 2.6 Login Page

Hotel Dashboard

New Booking

RoomsBookingsPayments

Room Management

View and manage all hotel rooms

Room Name	Type	Price (per night)	Status
Deluxe Suite	Suite	\$250	Available
Ocean View Room	Standard	\$150	Available
Presidential Suite	Suite	\$500	Occupied
Garden View Room	Standard	\$120	Available

Figure 2.7 Hotel Dashboard

Book a Room

Fill in the details to create a new reservation

Guest Name

Enter guest name

Select Room

Choose a room

Check-In Date

 Select date

Check-Out Date

 Select date

Confirm Booking

Figure 2.8 Booking Dashboard

2.9 Summary

This chapter has presented an analysis of the existing system and identified its major limitations. It also introduced the proposed system as a solution and defined both functional and non-functional requirements. Additionally, it provided an overview of system analysis to support the system design in the next chapter.

CHAPTER THREE

SYSTEM DESIGN

3.1 Introduction

This chapter presents the system design of the Hotel Booking System. System design is a crucial phase in software development that defines the architecture, components, modules, interfaces, and data flow. The main purpose of this chapter is to transform the system requirements into a structured solution that can be implemented efficiently.

The Hotel Booking System is designed to simplify the process of booking hotel rooms, manage reservations, and improve communication between customers and hotel administrators. This system ensures efficiency, accuracy, and a user-friendly interaction for both users and administrators.

3.2 System Architecture Design

The system follows a three-tier architecture consisting of the following layers:

3.2.1 Presentation Layer (User Interface)

This layer handles user interaction through the web interface and allows users to view rooms, book rooms, and manage their accounts.

- User login page
- Registration form
- Room booking interface
- Admin dashboard

The interface is designed to be simple, responsive, and user-friendly.

3.2.2 Application Layer (Business Logic)

This layer processes all core functionalities such as booking validation, payment processing, and room availability checking.

- User authentication
- Room availability checking
- Booking processing
- Payment handling logic
- Data validation

3.2.3 Data Layer (Database)

This layer is responsible for storing and retrieving data in the database, including user details, room information, bookings, and payments.

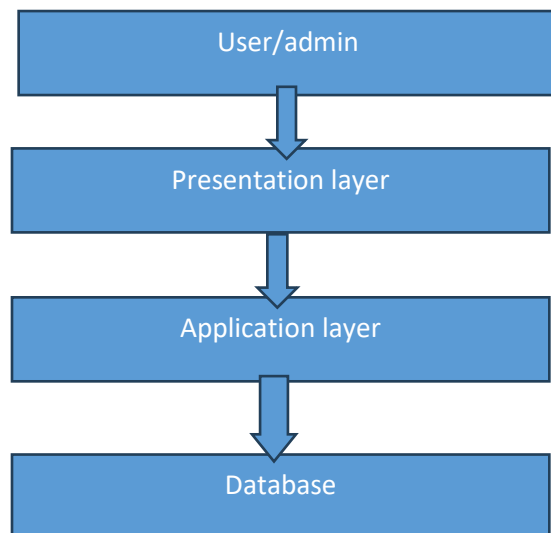
- User information
- Room details
- Booking records
- Payment records
- Admin data

The database ensures data consistency and security.

3.2.4 Architecture Diagram

The architecture diagram illustrates the three-tier structure of the system, including presentation layer, business logic layer, and database layer.

The presentation layer handles user interaction, the application layer processes business logic, and the data layer manages data storage and retrieval.



3.3 System Modules Design

The system is divided into different modules to improve maintainability and scalability.

3.3.1 User Module

This module allows users to:

- Register an account
- Log in to the system
- Search available rooms
- Book rooms
- Cancel a booking

- View booking history

3.3.2 Admin Module

This module is used by hotel administrators to:

- Manage rooms (add, update, delete)
- Manage bookings
- Manage users
- View reports
- Confirm or reject bookings

3.3.3 Booking Module

This module handles:

- Room reservation process
- Availability checking
- Booking confirmation
- Booking cancellation

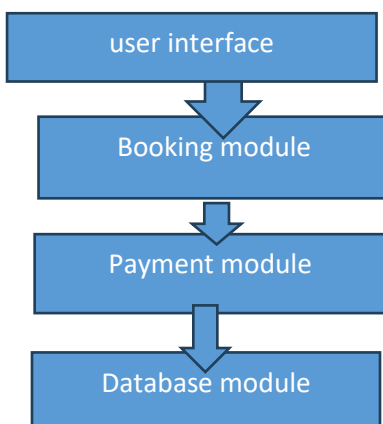
3.3.4 Payment Module

This module manages:

- Payment processing
- Payment confirmation
- Receipt generation

3.3.5 Component Diagram

The component diagram illustrates how different modules such as user management, booking, payment, and reports interact through defined interfaces within the system.



3.4 System Design Methodology

The system was designed using the Object-Oriented Design (OOD) approach. This methodology helps organize the system into classes and objects that interact with each other.

Key principles of OOD used in the system include:

- Encapsulation
- Modularity
- Reusability
- Maintainability
- Inheritance
- Abstraction

Examples of classes used in the system include:

- User
- Admin
- Room
- Booking
- Payment

3.5 Data Flow Design

The system data flow describes how information flows within the system.

3.5.1 Level 0 Data Flow Diagram (Context Diagram)

The system interacts with two main external entities:

- Customer
- Admin

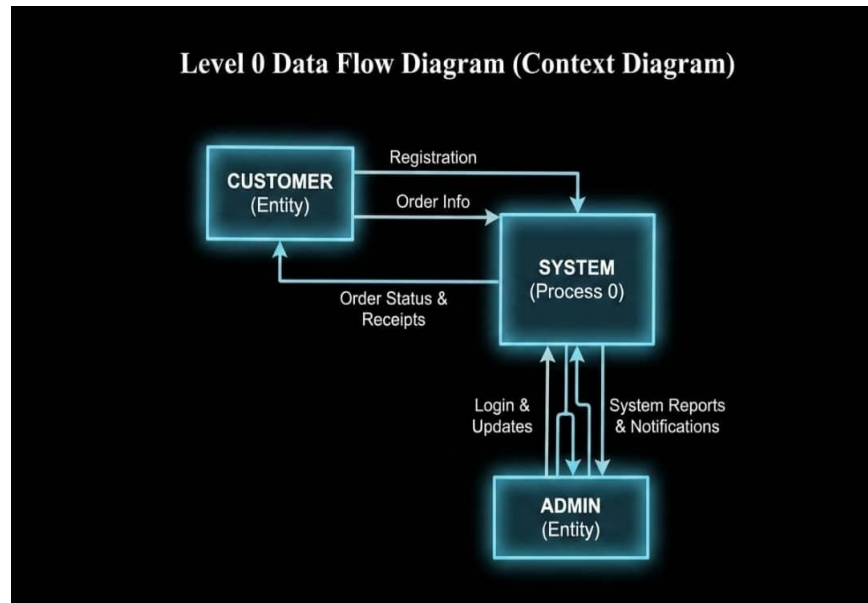


Figure 3.1 Level 0 Data Flow Diagram

Customers send booking requests, and the system responds with confirmations. The admin manages system data and receives reports.

3.5.2 Level 1 Data Flow Diagram

This includes:

- User authentication process
- Booking process
- Room management process
- Payment processing

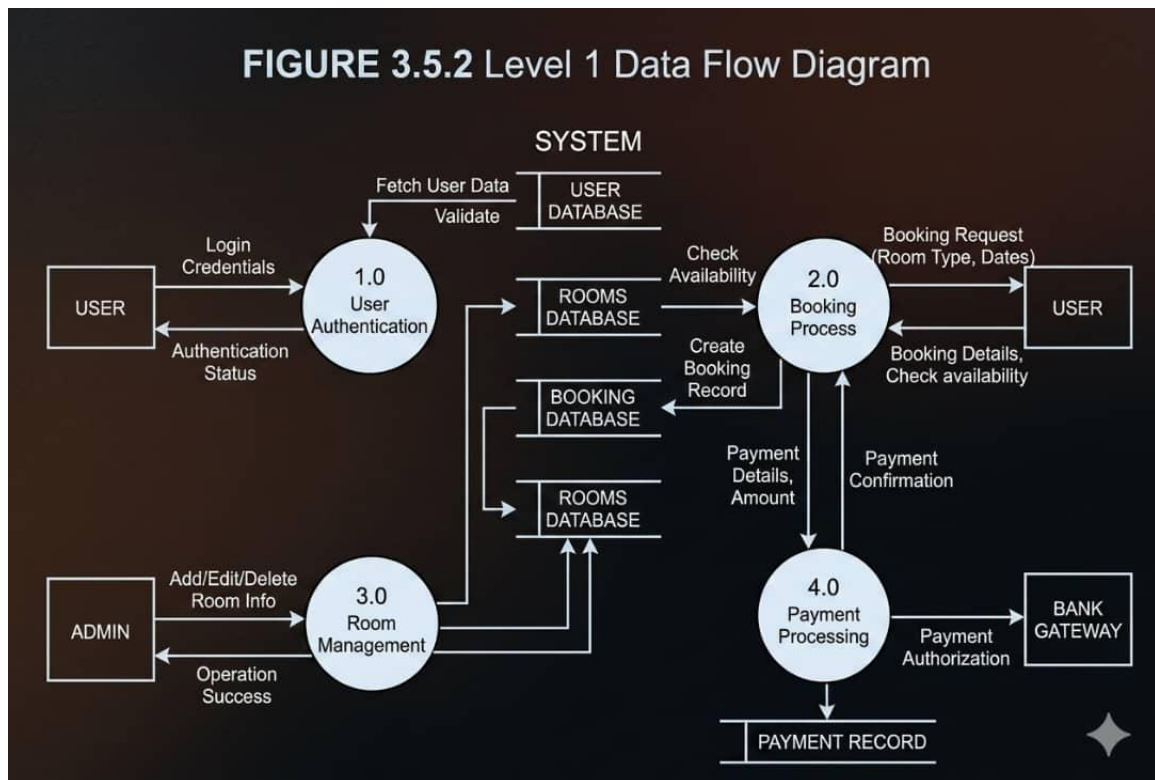


Figure 3.2 Level 1 Data Flow Diagram

3.6 Database Design

The system database is designed using relational database principles.

3.6.1 Main Tables

User Table

Field Name	Data Type	Key
UserID	INT	PK
FullName	VARCHAR	
Email	VARCHAR	
Phone	VARCHAR	
Password	VARCHAR	
Role	VARCHAR	
CreatedAt	DATETIME	

Table 3.1 Users Table

Table 3.1 Rooms Table

Rooms Table

Field Name	Data Type	Key
RoomID	INT	PK
RoomNumber	VARCHAR	
RoomType	VARCHAR	
Price	DECIMAL	
Status	VARCHAR	
Capacity	INT	

Table 3.2 Booking Table

Booking Table

Field Name	Data Type	Key
BookingID	INT	PK
UserID	INT	FK
RoomID	INT	FK
CheckInDate	DATE	
CheckOutDate	DATE	
TotalAmount	DECIMAL	
Status	VARCHAR	

Table 3.3 Payments Table

Payments Table

Field Name	Data Type	Key
PaymentID	INT	PK
BookingID	INT	FK
Amount	DECIMAL	
PaymentDate	DATETIME	
Method	VARCHAR	
Status	VARCHAR	

Table 3.4 Admin Table

Admin Table

Field Name	Data Type	Key
AdminID	INT	PK
Name	VARCHAR	
Email	VARCHAR	
Password	VARCHAR	

3.6.2 Entity Relationship Diagram

An Entity Relationship Diagram (ERD) is a graphical representation used to model the structure of a database system. It illustrates system entities, their attributes, and the relationships among them. ERDs are important during the database planning stage because they help developers understand how data is organized and connected before implementation.

In this system, entities such as User, Room, Booking, and Payment are connected through relationships that define how booking and payment processes are managed.

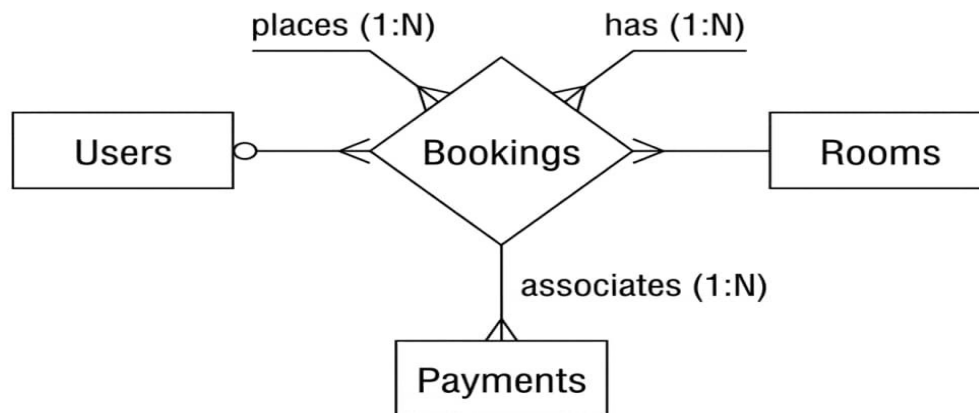


Figure 3.3 Entity Relationship Diagram

3.7 System Interface Design

The interface design focuses on usability and simplicity.

3.7.1 User Interface

- A home page displaying available rooms
- Search bar for rooms
- Booking form
- Login and registration pages

3.7.2 Admin Interface

- Dashboard showing statistics
- Room management panel
- Booking management section
- Reports section

3.8 Security Design

Security is an important part of the system. The following security measures are implemented:

- Password hashing using BCrypt
- User authentication and authorization
- Role-based access control
- Secure access to the database
- Session management
- Input validation to prevent SQL injection attacks
- HTTPS secure connection

3.9 System Requirements

3.9.1 Hardware Requirements

- Processor: Intel Core i3 or higher recommended
- RAM: 4GB minimum
- Storage: 500GB HDD or higher

3.9.2 Software Requirements

Operating System: Windows or Linux

Database: MySQL

- Programming Language: PHP
- Web Browsers: Google Chrome and Mozilla Firefox

3.9.3 Deployment Diagram

The deployment diagram explains how the system is deployed on a server, client devices, and a connected database.

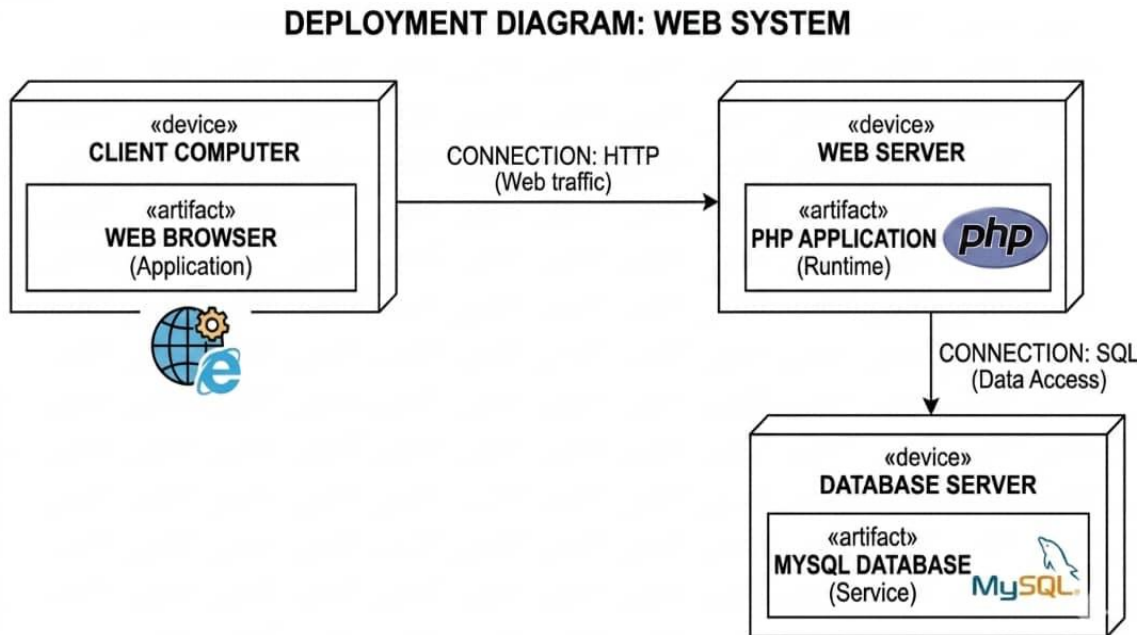


Figure 3.4 Deployment Diagram

3.10 Summary

This chapter has presented the overall system design of the Hotel Booking System. It explained the system architecture, modules, database design, interface design, and security mechanisms. The design ensures that the system is efficient, scalable, secure, and easy to use. The next chapter will focus on system implementation and testing.